# Simulation

# Things to remember!

- Simulation is different what we have learnt so far such including linear programing, queuing theory, decision analysis etc.
- We use simulation when systems are very complex, uncertain and dynamic
- We will learn the logic of simulation here with some examples.

# Simple question

- How many tellers should a bank assign, so customers don’t wait too long, but the bank also don’t waste salary cost?
- Two questions can help us to solve this problem.
  - Can we solve this exactly?
  - Or should we reproduce the customer arrivals and service for a certain period of time (a day, a month, a quarter etc?
- This is where we use simulations

# Introduction to Simulation in Operations Research

- Simulation involves using a computer to imitate/mimic (simulate) the operation of an entire process or system.
  - Computer builds a virtual copy of the system
- Helps to understand the behavior of the system – observes what happens
- Use results for decision making
- In case operation research simulation may be taken as a mathematical or statistical model to understand the behavior of the decision making in future rather decision variables.
- It is one of the most widely used techniques in operations research.
- Flexible, powerful, and intuitive, leading to rapid growth in popularity.

# Simulation

- Simulation process
  - Copy the Real World System
  - Model it
  - Conduct Computer Experiment
  - Results
  - Decision
- Simulation is not just “guessing”
- It is a structured experiment
- It is especially useful when randomness is involved

# What Kind of Systems Are Simulated?

**Stochastic Systems:** A system that evolves probabilistically over time.

Stochastic means random or uncertain behavior.

**Examples:** Number of customers in a queue; Vehicles at a toll plaza; Patients arriving at a clinic; Water demand in irrigation canals; Prices in uncertain markets

**Key concept**: Simulation is especially useful when the future depends on chance events.

# How Simulation Works Conceptually

1. Describe the system
2. Identify what can happen
3. Represent randomness with probability
4. Let the system “run” on a computer
5. Record performance results
6. Compare alternatives

# A Simple Example

- For a bank:
  - Customers arrive randomly
  - Tellers serve randomly
  - Waiting time is measured
- **Input randomness → Events → State changes → Output performance**

# Requirement of Simulation

- We must have some past data collected
- We have to present the data in frequency distribution
- Probability distribution
- Cumulative probability distribution
- Determine random number classes/ interval
- Getting the number relevant to the decision variable
- Understanding the behavior of decision variable under the random occurring phenomena.

# Key Characteristics

- Mimics randomness and dynamic behavior.
- Used when analytical solutions are impractical.
- **Example:**
  - Simulating customer arrivals at a bank to optimize teller staffing.
  - Flight simulator for pilot training
  - Traffic simulation for roads
  - Hospital patient flow simulation
  - Supply chain simulation
  - Agricultural water allocation simulation

# Why Do We Need Simulation?

- Why Not Just Solve the Problem Mathematically?
- Because many real systems are:
  - Complex
  - Random
  - Dynamic over time
  - Interconnected
  - Difficult to model exactly

**Examples:** Customer arrivals are uncertain ; Machine failures happen randomly ; Traffic congestion changes by minute

# Why Use Simulation? (Advantages)

- Flexibility:
  - Simulation models complex systems (e.g., queues, inventory, supply chains).
- Risk-Free Testing:
  - Evaluates "what-if" scenarios without real-world costs.
- Time Compression:
  - Simulates years of operation in seconds.

# Key Components of a Simulation Model

- **System State:** Variables (e.g., # of customers in line).
- **Events:** Actions that change the state (e.g., arrivals, service completions).
- **Clock:** Tracks simulated time.
- **Randomness:** Generates stochastic events (e.g., using random numbers).
- **Rules:** Logic for state transitions (e.g., "If arrival, increment queue").
- **Performance Metrics:** Outputs (e.g., average wait time).

# Example

**Bank queue**

- State = number of customers
- Event = arrival/departure
- Clock = 9:00, 9:03, 9:07…
- Randomness = arrival/service times
- Transition = if arrival occurs, queue +1

# Types of Simulation

- **Discrete-Event Simulation (DES):**
  - Events occur at specific point of times (e.g., customer arrivals, customer leaves, machine breaks).
- **Continuous Simulation:**
  - State changes continuously over time in a smooth way (e.g., fluid flow, temperature).
  - Example: Aircraft flight path.

# Practical Challenges

- Computational Cost: Long runs for precision.
- Data Requirement: Requires accurate input distributions.
- Statistical Analysis: Outputs are estimates, not exact.

# Real-World Applications

- Healthcare: Patient flow.
- Manufacturing: Production line optimization.
- Finance: Portfolio risk analysis.

# Common Types of Simulation

# Queueing Systems

- Call centers, banks, transportation hubs.
- Example:
  - Bank Teller Optimization: Simulate customer arrivals/service times to minimize wait times.

# Distribution & Logistics

- Warehouse design, vehicle routing, supply chain disruptions.
- Simulation Benefits:
- Evaluates trade-offs (cost vs. service level).

# Financial Risk Analysis

- Portfolio risk (VaR (value at Risk) calculations).
- Option pricing.
- Loan default modeling.
- Method:
  - Generate thousands of market scenarios.

# How the Coin Game Becomes a Simulation

- **Manual simulation:**
  - Flip a real coin many times
  - Record number of flips
- **Computer simulation:**
  - Generate random numbers from **0 to 1**
  - Use:
  - **0.0000 – 0.4999 ➔ Heads**
  - **0.5000 – 0.9999 ➔Tails**

# Outputs of a Simulation

- Simulation helps us estimate:
  - Average waiting time
  - Average profit or loss
  - Probability of congestion
  - Resource utilization
  - Risk of bad outcomes
  - Etc.

# Simulation vs Analytical Methods

**Use analytical methods when:**

- Problem is mathematically manageable
- Exact solution is possible
- Sensitivity analysis is needed quickly

**Use simulation when:**

- System is too complex
- Randomness is essential
- Real-world detail matters
- We want to compare policies before implementation

# Simulation Applications

- Major applications include:
  - queueing
  - inventory
  - projects
  - Manufacturing
  - distribution
  - finance
  - health care
  - service systems
  - military systems

# Generation of Random Numbers

# Why Random Numbers Matter

- Simulations rely on randomness to model uncertainty.
- Used for:
  - Event timing (e.g., arrivals).
  - Sampling from distributions.
- Example:
- Coin flips in games, service times in queues.

# How Does a Computer Flip a Coin?

# Random Number Generator

- A random number generator (RNG) is an algorithm that produces a sequence of numbers that follow a specified probability distribution and appear random.

# True Random vs Pseudo-Random

- Are Computer Random Numbers Really Random?
- Not exactly. The computer-generated random numbers are technically not truly random, because they are predictable and reproducible if we know the generator and starting value.
- Therefore, they are often called Pseudo-random numbers
- But they are still useful if:
  - they look random,
  - they are well distributed,
  - and they are statistically acceptable.

# Desired Properties of Random Numbers

- Uniformity: Evenly distributed over [0, 1].
- Independence: No correlation between numbers.
- Long Period: Repeats only after many numbers.

# Types of Random Numbers

**Random Integer Numbers**

- Examples:
- 0, 1, 2, ..., 9
- 00, 01, 02, ..., 99
- 0 to 31
- 0 to 999

**Uniform Random Numbers**

- These are decimal values uniformly spread over: **[0, 1]**
- Examples:
- 0.1273
- 0.5834
- 0.9410

# How Random Numbers Drive Simulation

- To simulate a fair coin:
- **Assign:**
  - 0.0000 to 0.4999 → Heads
  - 0.5000 to 0.9999 → Tails
- This is exactly how the textbook illustrates coin-flip simulation.
- **Excel formula:** =IF(A1<0.5,"Heads","Tails")

| Random Number | Result |
|---|---|
| **0.3039** | Heads |
| **0.7914** | Tails |
| **0.4250** | Heads |

# Random Numbers in Excel

| RAND() | RANDBETWEEN(1, 100) | RAND()*100 | RANDARRAY(5,1,1,100,TRUE) | RANDARRAY(5,1,1,100) |
|---|---|---|---|---|
| 0.738516 | 32 | 81.1199774 | 17 | 81.04199365 |
| 0.328236 | 20 | 56.7109452 | 74 | 91.32555997 |
| 0.176959 | 87 | 64.9747497 | 46 | 32.08487987 |
| 0.065912 | 98 | 51.1306948 | 45 | 50.16262537 |
| 0.135566 | 6 | 64.2111389 | 41 | 95.87849386 |
| 0.62256 | 60 | 51.0432651 | 8 | 71.4960532 |
| 0.582513 | 54 | 73.0550288 | 57 | 8.690555121 |
| 0.175442 | 73 | 59.7845868 | 72 | 82.43114855 |
| 0.891907 | 49 | 34.7663769 | 39 | 45.48286048 |
| 0.478816 | 12 | 92.4637712 | 71 | 28.29115253 |
| 0.443645 | 75 | 39.9747025 | 75 | 76.18036361 |
| 0.131649 | 77 | 41.4232072 | 76 | 59.91589332 |
| 0.077857 | 32 | 5.83771544 | 100 | 71.59390432 |
| 0.449036 | 75 | 29.8892603 | 29 | 19.53647623 |
| 0.630572 | 75 | 56.6612049 | 100 | 87.65827028 |
| 0.998509 | 88 | 31.9809906 | 23 | 85.35269866 |

# How Computers Generate Random Integer Numbers

- **Common method is Congruential methods**
  - additive congruential method
  - multiplicative congruential method
  - mixed congruential method
- **Mixed Congruential Method is the most important one**
  - This is the mathematical engine inside many random number generators.

# Mixed Congruential Method

- $Algorithm$
- $x_{n+1} = (ax_n + c) \mod m$
- $a = multiplier, c = increatment,\ m = modulas$
- Example:
- $a = 5, c = 7, m = 8, seed(x_0 = 4) \rightarrow Sequence = 3, 6, 5, 0, 7\ , 2\ , 1.$

It means take:

1. Take the current number
2. multiply by a
3. add c
4. divide by m
5. keep the remainder

# Mixed Congruential Method – A simple step by step explanation of the previous slides.

- The MCM generates pseudo-random numbers using a simple recursive formula.
- Step 1. Start with a seed ($x_0$)
  - The $x_0$ is the initial value we choose manually ➔the seed value
- Step 2. Calculate $x_1$

  $x_1 = (a.x_0 + c)$ mod $m == (5\,.\,4 + 7)\ mod\ 8$

  27 mod 8 ➔ 27% 8 with a remainder of 3.

  So $x_1 = 3$
- Step 3. Calculate $x_2$
- Now instead of $x_0$, you will use $x_1$ value.

- Through this way the MCM deterministically generates numbers that appear random.
- The sequence repeats after a full period

# Example Calculation

$Multipliar\ (a) = 5;$
Increment $(c) = 7;$
$modulas\ (m) = 8;$
$seed(x_0 = 4)$

■ **TABLE 20.4** Illustration of the mixed congruential method

| $n$ | $x_n$ | $5x_n + 7$ | $(5x_n + 7)/8$ | $x_{n+1}$ |
|---|---|---|---|---|
| 0 | 4 | 27 | $3 + \frac{3}{8}$ | 3 |
| 1 | 3 | 22 | $2 + \frac{6}{8}$ | 6 |
| 2 | 6 | 37 | $4 + \frac{5}{8}$ | 5 |
| 3 | 5 | 32 | $4 + \frac{0}{8}$ | 0 |
| 4 | 0 | 7 | $0 + \frac{7}{8}$ | 7 |
| 5 | 7 | 42 | $5 + \frac{2}{8}$ | 2 |
| 6 | 2 | 17 | $2 + \frac{1}{8}$ | 1 |
| 7 | 1 | 12 | $1 + \frac{4}{8}$ | 4 |

# The Cycle Length

- The number of consecutive numbers in a sequence before it begins repeating itself is referred to as the cycle length.
- Thus, the cycle length in the example is 8.
- The maximum cycle length is m, so the only values of a and c considered are those that yield this maximum cycle length.
- A good generator should have a long cycle length.

# Converting Random integers to Uniform Random Numbers

- $Uniform\ RN = \frac{Random\ Integer + 1/2}{m}$
- Example from previous slide
- $x_n = 3 \rightarrow \frac{3+0.5}{8} = 0.4375$
- $x_n = 6 \rightarrow \frac{6+0.5}{8} = 0.8125$

**TABLE 20.5** Converting random integer numbers to uniform random numbers

| Random Integer Number | Uniform Random Number |
|---|---|
| 3 | 0.4375 |
| 6 | 0.8125 |
| 5 | 0.6875 |
| 0 | 0.0625 |
| 7 | 0.9375 |
| 2 | 0.3125 |
| 1 | 0.1875 |
| 4 | 0.5625 |

# Simulation

- A technique involves conducting repetitive experiments on the model of the system under study, with some known probability distribution to draw random samples (observations) using random numbers.
  - Setting up a probability distribution for variables to be analyzed.
  - Building a cumulative probability distribution for each random variable.
  - Generating random numbers and then assigning an appropriate set of random numbers to represent value or range (interval) of values for each random variable.
  - Conducting the simulation experiment using random sampling.
  - Repeating Step – 4 until the required number of simulation runs has been generated.
  - Designing and implementing a course of action and maintaining control.

# Example

- Dr. Paul is a well-known dentist who schedules all his patients for 30-minutes appointments.
- Some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probabilities and the time actually needed to complete that work:
- Category: Filling Crown Cleaning Extraction Check-up
- Time(Min): 45 60 15 45 15
- Probability: 0.40 0.15 0.15 0.10 0.20
- Simulate the dentist's clinic for 4 hours and determine average waiting time for the patients as well as the idleness of the doctor.
- Assume that all the patients show up at the clinic at exactly their scheduled arrival time starting at 8AM. Use the following random numbers to solve the above problem: 40, 82, 11, 34, 25, 66, 17, 79.

# Step 1: Given information

- Time frame: 4 hours = 240 minutes

  Appointment interval: Every 30 minutes → Total patients = 240 / 30 ➔ 8 patients

  Random numbers provided: 32, 82, 11, 34, 25, 66, 17, 79

# Step 2: Categories and Their Probabilities

- Probability is given here. If it is not given, the frequency must be given and on the basis of the frequency we can calculate probability ➔ $prob = f / \sum f$

| Category | Time (min) | Probability | Cumulative Probability | Range |
|---|---|---|---|---|
| Filling | 45 | 0.40 | 0.40 | 01–40 |
| Crown | 60 | 0.15 | 0.55 | 41–55 |
| Cleaning | 15 | 0.15 | 0.70 | 56–70 |
| Extraction | 45 | 0.10 | 0.80 | 71–80 |
| Check-up | 15 | 0.20 | 1 | 81–100 |

# Converting probabilities into number intervals

- **Filling = 0.40** ➔ $0.40 \times 100 = 40$ =➔ Filling gets **40 random numbers**: $00 - 39$
- **Crown = 0.15** ➔ $0.15 \times 100 = 15$ =➔ Crown gets the **next 15 numbers**: $40 - 54$
- **Cleaning = 0.15** ➔ $0.15 \times 100 = 15$ =➔ Cleaning gets: $55 - 69$
- **Extraction = 0.10** ➔ $0.10 \times 100 = 10$ =➔ Extraction gets: $70 - 79$
- **Check-up = 0.20** ➔ $0.20 \times 100 = 20$ =➔ Check-up gets: $80$ to $99$

# Step 3: Simulation Table

- Random number are given, these numbers need to be matched with the range in previous slide to understand the service category and thus service time.

| Patient | Random No. | Category | Service Time | Arrival Time | Start Time | Waiting Time | End Time | Doctor Idle Time |
|---|---|---|---|---|---|---|---|---|
| 1 | 32 | Filling | 45 | 8:00 AM | 8:00 AM | 0 | 8:45 AM | 0 |
| 2 | 82 | Check-up | 15 | 8:30 AM | 8:45 AM | 15 | 9:00 AM | 0 |
| 3 | 11 | Filling | 45 | 9:00 AM | 9:00 AM | 0 | 9:45 AM | 0 |
| 4 | 34 | Filling | 45 | 9:30 AM | 9:45 AM | 15 | 10:30 AM | 0 |
| 5 | 25 | Filling | 45 | 10:00 AM | 10:30 AM | 30 | 11:15 AM | 0 |
| 6 | 66 | Cleaning | 15 | 10:30 AM | 11:15 AM | 45 | 11:30 AM | 0 |
| 7 | 17 | Filling | 45 | 11:00 AM | 11:30 AM | 30 | 12:15 PM | 0 |
| 8 | 79 | Extraction | 45 | 11:30 AM | 12:15 PM | 45 | 1:00 PM | 0 |

# Results

Total Waiting Time:

= 15 + 0 + 15 + 30 + 45 + 30 + 45 = 180 minutes

Average Waiting Time:
= 180 / 8 = 22.5 minutes

- Total Doctor Idle Time:
  = 0 minutes (since the doctor never finishes early before the next patient)

Thanks